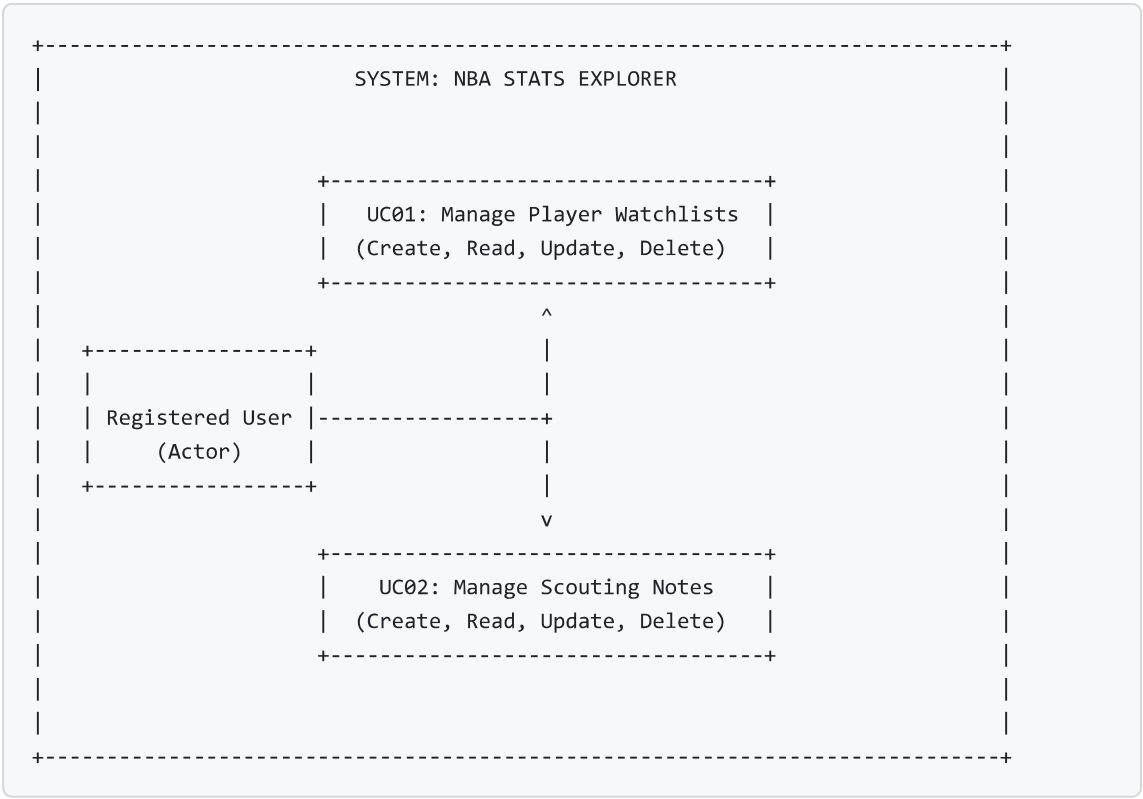
Systems Analysis & Design Specification

Project: NBA Stats Explorer

Module: Player Watchlists & Scouting Notes CRUD

Points: 10 PTS

Deliverable 1: Use Case Model



Deliverable 2: Use Case Documents

Use Case Document 1: UC01 - Manage Player Watchlists

Use Case Field	Description / Value
Use Case ID	**UC01**
Use Case Name	Manage Player Watchlists
Primary Actor	Registered User
Goal	Create, view, update, and remove personalized collections of NBA players to track performance statistics.
Trigger	User navigates to the "My Watchlists" section from the application navigation bar.
Preconditions	User is authenticated (signed in) and the watchlists dashboard is accessible (`watchlists.php`).
Success Outcome	System validates input, persists/updates the watchlist selection, displays success status, and renders list details.
Failure Outcome	System keeps the form visible, highlights specific validation errors, preserves valid field inputs, and leaves NBA entity data untouched.

Main Success Scenario (Normal Flow)

1. User clicks **"+ Create New Watchlist"** (`watchlist-create.php`) or selects **"Edit"** on an existing watchlist (`watchlist-edit.php`).
2. System displays the Watchlist Form with fields: *Watchlist Name* (text), *Description* (textarea), and *Player Selection* (checkboxes).
3. User enters a unique Watchlist Name, an optional description, selects at least one player, and submits the form.
4. System validates server-side: verifies Name is not empty, checks length constraints (max 100 chars), and ensures at least one valid player is checked.
5. System accepts the intake/update, displays a confirmation message, and redirects the user to the Watchlist Dashboard (`watchlists.php`).
6. User can view saved lists or click the inline **"Delete"** button to remove a list after confirming the prompt.

Alternative & Exception Flows (Failure Paths)

- **Alt Flow 3a: Missing Watchlist Name**
  - System flags `Name` field as required, retains checked players, and displays `"Watchlist name is required."`.
- **Alt Flow 3b: No Players Selected**
  - System detects zero checked players and displays `"At least one player must be selected."`.
- **Alt Flow 6a: Deletion Cancellation**
  - User cancels the inline deletion confirmation modal; system leaves the watchlist intact.

Use Case Document 2: UC02 - Manage Scouting Notes

Use Case Field	Description / Value
Use Case ID	**UC02**
Use Case Name	Manage Qualitative Scouting & Analysis Notes
Primary Actor	Registered User
Goal	Create, view, update, and remove detailed qualitative scouting notes, category tags, and impact ratings for individual players.
Trigger	User clicks "Scouting Notes" in the navigation header or selects "Add Note" on a player card.
Preconditions	User is signed in and the Scouting Notes dashboard is accessible (`scouting-notes.php`).
Success Outcome	System validates note content and metadata, saves/updates the record, and displays the updated notes board.
Failure Outcome	System prevents submission, highlights failed field rules (e.g., character overflow), and keeps input text intact.

Main Success Scenario (Normal Flow)

1. User selects **"+ Add New Scouting Note"** (`scouting-note-create.php`) or clicks **"Edit Note"** (`scouting-note-edit.php`).

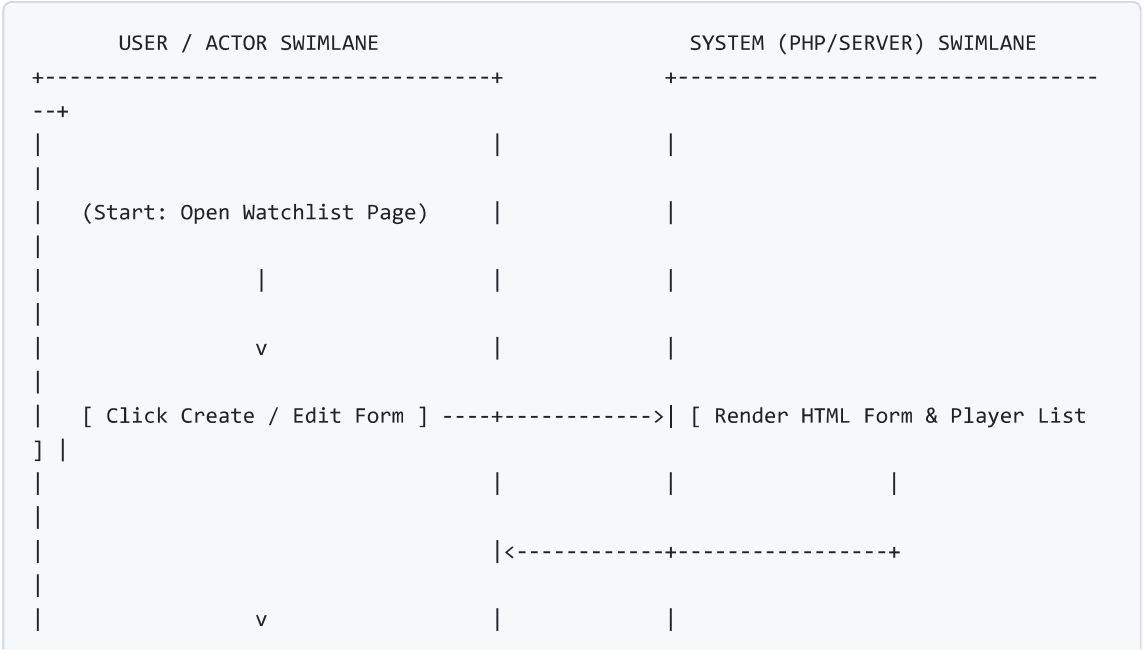
2. System loads the form containing: *Note Title*, *Target Player/Team*, *Category Tag* (e.g., Breakout Candidate, Trade Target), *Impact Rating* (1–5 Stars), and *Analysis Content* (textarea).
3. User enters a title, chooses a target player, selects a rating and tag, enters detailed notes, and clicks **"Save Note"**.
4. System validates server-side: ensures Title and Player are present, Rating is within range (1-5), and Content does not exceed 1,000 characters.
5. System confirms entry, redirects to `scouting-notes.php`, and displays the note card on the dashboard.
6. User can browse existing notes or trigger an inline **"Delete"** button to remove an entry.

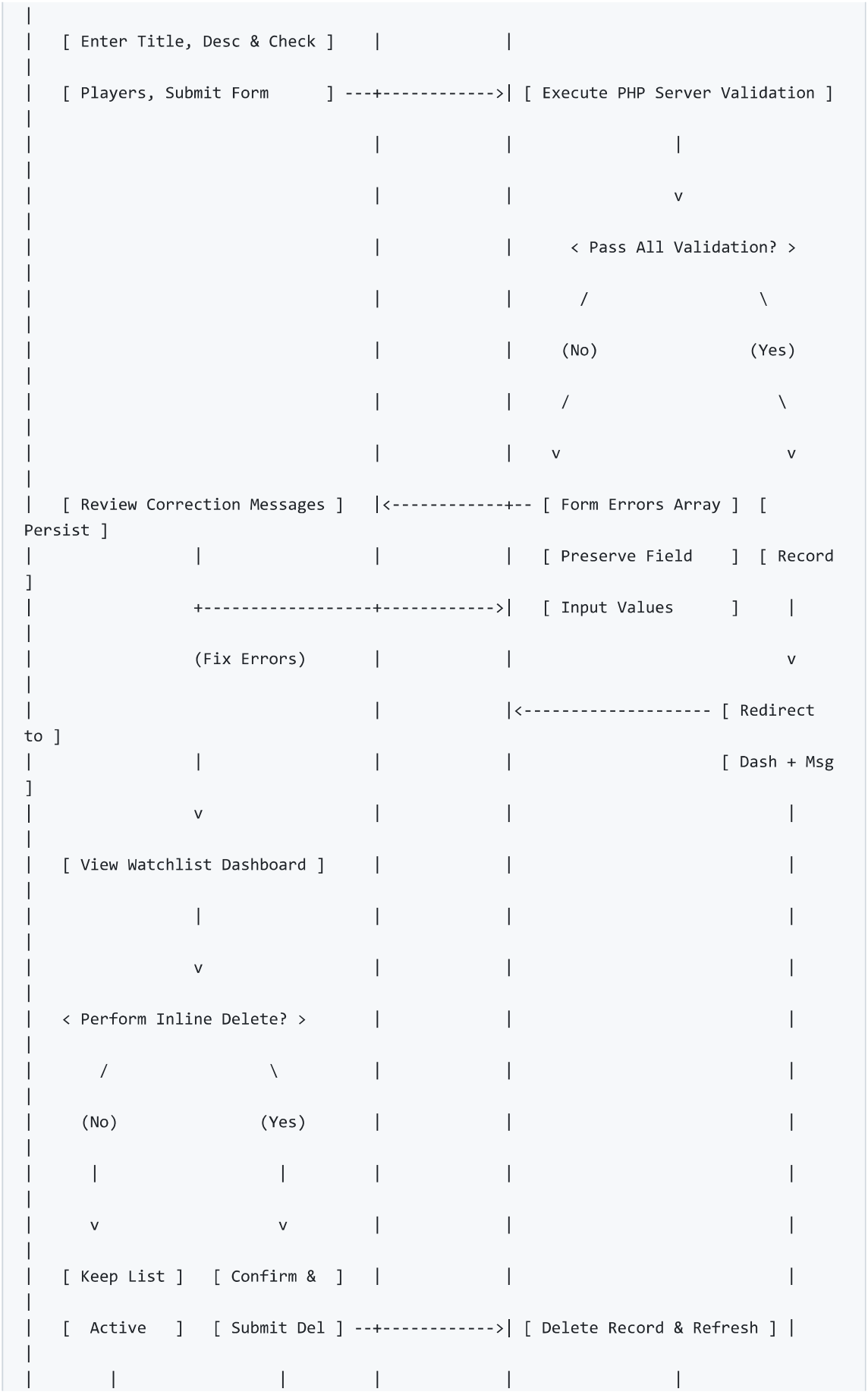
Alternative & Exception Flows (Failure Paths)

- **Alt Flow 3a: Missing Note Title or Target Player**
  - System alerts `"Title and Target Player are required fields."`.
- **Alt Flow 3b: Text Content Exceeds Character Limit (> 1000 characters)**
  - System rejects submission and displays `"Analysis content cannot exceed 1000 characters."`.

Deliverable 3: Use Case Activity Diagrams

Activity Diagram 1: UC01 - Manage Player Watchlists

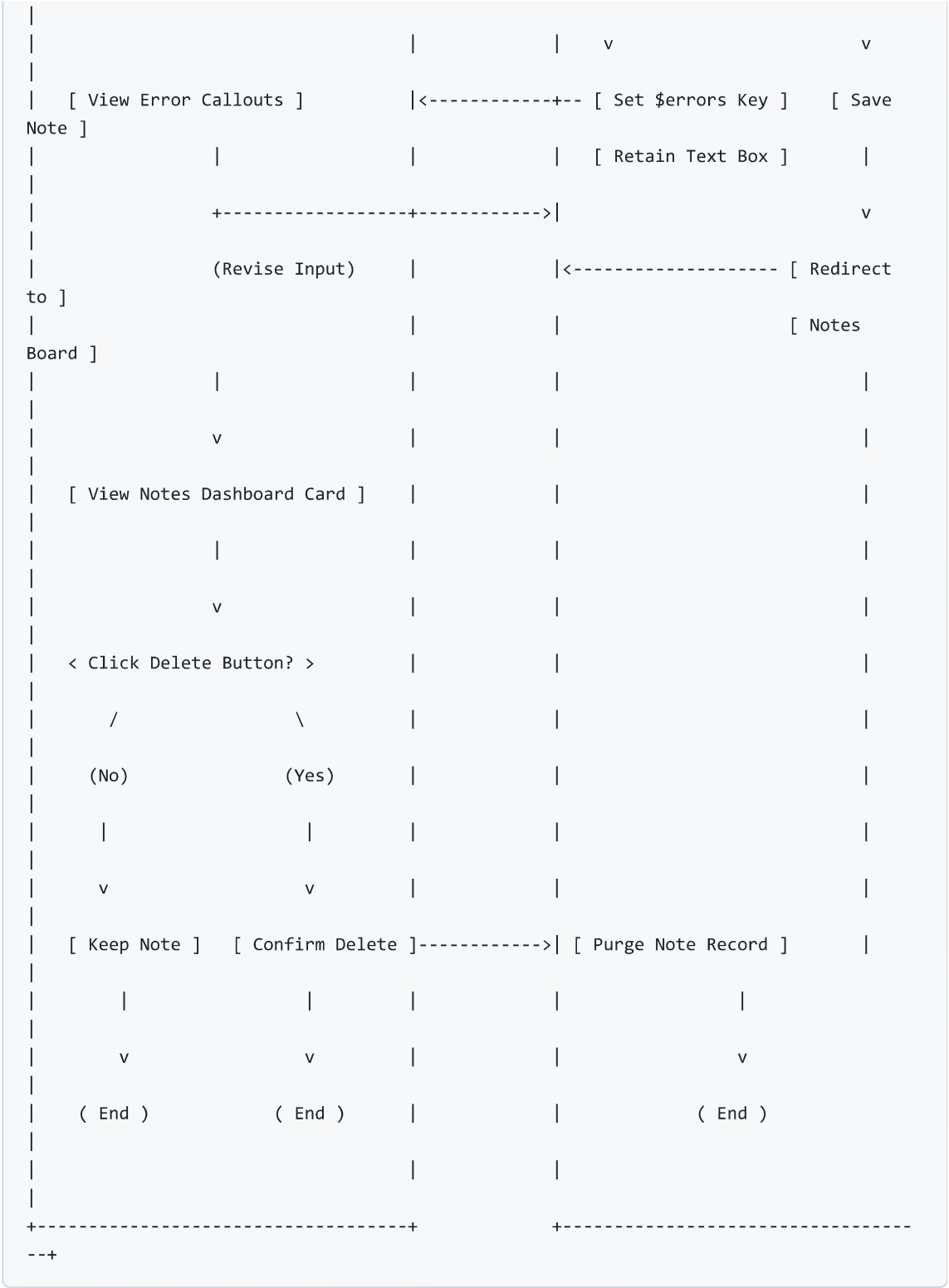



[Enter Title, Desc & Check]
[Players, Submit Form]
[Execute PHP Server Validation]
< Pass All Validation? >
(No)
(Yes)
[Review Correction Messages]
[Form Errors Array]
[Persist]
[Preserve Field]
[Record]
[Input Values]
(Fix Errors)
[Redirect to]
[Dash + Msg]
[View Watchlist Dashboard]
< Perform Inline Delete? >
(No)
(Yes)
[Keep List]
[Active]
[Confirm & Submit Del]
[Delete Record & Refresh]

```
|
|       v                v         |             |                v
|
|    ( End )            ( End )      |             |            ( End )
|
|                                    |             |
|
+------------------------------------+             +--------------------------------
--+
```

Activity Diagram 2: UC02 - Manage Scouting Notes

```
       USER / ACTOR SWIMLANE                         SYSTEM (PHP/SERVER) SWIMLANE
+-----------------------------------+             +--------------------------------
--+
|                                   |             |
|
|   (Start: Open Scouting Notes)    |             |
|
|                |                  |             |
|
|                v                  |             |
|
|   [ Click Create Note Button ] ----+------------>| [ Render Scouting Form & Fields ]
|
|                                   |             |                |
|
|                                  |<------------+-----------------+
|
|                v                  |             |
|
|   [ Enter Title, Select Player, ]  |             |
|
|   [ Rating, Tag & Analysis Text ] -+------------>| [ Execute PHP Server Rules ]
|
|                                   |             |   - Required Fields Check
|
|                                   |             |   - Max Length <= 1000 chars
|
|                                   |             |                |
|
|                                   |             |                v
|
|                                   |             |      < Pass All Validation? >
|
|                                   |             |       /                 \
|
|                                   |             |     (No)               (Yes)
|
|                                   |             |     /                     \
```


[View Error Callouts]
[Set $errors Key]
[Save Note]
[Retain Text Box]
(Revise Input)
[Redirect to]
[Notes Board]
[View Notes Dashboard Card]
< Click Delete Button? >
(No)
(Yes)
[Keep Note]
[Confirm Delete]
[Purge Note Record]
(End)
(End)
(End)